

Homework 5
CSDA5110
Fall 1, 2024

- 1) There was a study of the relation between degree of brand liking (Y) and moisture content (X_1) and sweetness (X_2) of the product. The data is on Canvas.
 - a. Obtain the scatter plot matrix and the correlation matrix. What information do these diagnostic aids provide here?
 - b. Fit the regression model to the data. State the estimated regression function. Interpret b_1 .
 - c. Obtain the residuals and prepare boxplot. What information does it provide?
 - d. Plot the residuals against the fitted values, X_1 and X_2 . Interpret them.
 - e. Prepare a normal probability plot of the residuals and obtain the test statistics. State the results.
 - f. Test whether there is a regression relation, using $\alpha = .01$. State the alternatives, decision rule, and conclusion. What does your test imply about β_1 and β_2 ?
 - g. Calculate the coefficient of multiple determination R^2 . How is it interpreted here?
 - h. Obtain an interval estimate of $E\{Y_{x0}\}$ when $X_{10} = 5$ and $X_{20} = 4$. Use a 99 percent confidence coefficient. Interpret your interval estimate.
 - i. Obtain a prediction interval for a new observation $Y_{0(\text{new})}$ when $X_{10} = 5$ and $X_{20} =$ Use a 99 percent confidence coefficient.
- 2) The lengths of pregnancies in a small rural village are normally distributed with a mean of 265 days and a standard deviation of 15 days.
 - a. In what range would you expect to find the middle 50% of most pregnancies?
 - b. What proportion of pregnancies will be more than 275 days?
- 3) A certain car model has a mean gas mileage of 34 miles per gallon (mpg) with a standard deviation 5mpg. If a car is selected at random,
 - a) What is the probability that the mileage of the selected car is greater than 33 mpg?
 - b) What is the probability that the mileage of the car is between 33mpg and 35mpg?